Output:

Name: Ravi

Balance: 25000.0

Experiment 11: Understanding static

Program 11.1: Static Variable

```
class Student {
    String name;
    static int count = 0;
    Student(String name) {
        this.name = name;
        count++;
    }
    void display() {
        System.out.println("Name: " + name);
    }
}
public class StaticVariable {
    public static void main(String[] args) {
        Student s1 = new Student("Raina");
        Student s2 = new Student("Rinku");
        Student s3 = new Student("Amar");
        s1.display();
        s2.display();
        s3.display();
        System.out.println("Total Students: " + Student.count);
    }
}
```

Output:

Name: Raina

Name: Rinku

Name: Amar

Total Students: 3

Program 11.2: Static Methods

```
class Utility {
    static int square(int n) {
        return n * n;
    }
    static int cube(int n) {
        return n * n * n;
    }
    static long factorial(int n) {
        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact *= i;
        }
        return fact;
    }
}
public class StaticMethods {
    public static void main(String[] args) {
        int n = 5;
        System.out.println("Square: " + Utility.square(n));
        System.out.println("Cube: " + Utility.cube(n));
        System.out.println("Factorial: " + Utility.factorial(n));
    }
}
```

Output:

Square: 25

Cube: 125

Factorial: 120

Experiment 12: Introducing final

Program 12.1: Final Keyword

```
class Parent {
    final int VALUE = 100;
    final void display() {
        System.out.println("This is a final method.");
        System.out.println("Final variable: " + VALUE);
    }
}
final class FinalClass {
    void show() {
        System.out.println("This is a final class.");
    }
}
public class FinalDemo {
    public static void main(String[] args) {
        Parent p = new Parent();
        p.display();
        FinalClass f = new FinalClass();
        f.show();
    }
}
```

Output:

This is a final method.

Final variable: 100

This is a final class.

Program 12.2: Blank Final Variable

```
class Employee {
    final int employeeId;
    String name;
    Employee(int employeeId, String name) {
        this.employeeId = employeeId;
        this.name = name;
    }
    void display() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Name: " + name);
    }
}
public class BlankFinal {
    public static void main(String[] args) {
        Employee e1 = new Employee(101, "Raina");
        Employee e2 = new Employee(102, "Rinku");
        e1.display();
        System.out.println();
        e2.display();
    }
}
```

Output:

Employee ID: 101

Name: Raina

Employee ID: 102

Name: Rinku

Experiment 13:

Program 13.1: College and Department

```
class College {
    static class Department {
        String name;
        Department(String name) {
            this.name = name;
        }
        void display() {
            System.out.println("Department: " + name);
        }
    }
}
public class CollegeDemo {
    public static void main(String[] args) {
        College.Department d =
                new College.Department("CSE-AIML");
        d.display();
    }
}
```

Output:

Department: CSE-AIML

Program 13.2: Employee Address

```
class Employee {
    int id;
```

```
    String name;
    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }
    static class Address {
        String city;
        String state;
        Address(String city, String state) {
            this.city = city;
            this.state = state;
        }
        void display() {
            System.out.println("City: " + city);
            System.out.println("State: " + state);
        }
    }
}
public class EmployeeAddress {
    public static void main(String[] args) {
        Employee e = new Employee(101, "Ravi");
        Employee.Address address =
                new Employee.Address("Vijayawada", "Andhra Pradesh");
        System.out.println("Employee ID: " + e.id);
        System.out.println("Name: " + e.name);
        address.display();
    }
}
```

Output:

Employee ID: 101

Name: Ravi

City: Vijayawada

State: Andhra Pradesh

Experiment 14: Inner Classes

Program 14.1: Student Address

```
class Student {
    String name;
    int rollNo;
    Student(String name, int rollNo) {
        this.name = name;
        this.rollNo = rollNo;
    }
    class Address {
        String city;
        String state;
        Address(String city, String state) {
            this.city = city;
            this.state = state;
        }
        void display() {
            System.out.println("Student Name: " + name);
            System.out.println("Roll Number: " + rollNo);
            System.out.println("City: " + city);
            System.out.println("State: " + state);
        }
    }
}
```

```
public class StudentAddress {
    public static void main(String[] args) {
        Student s = new Student("Ravi", 101);
        Student.Address a =
                s.new Address("Vijayawada", "Andhra Pradesh");
        a.display();
    }
}
```

Output:

Student Name: Ravi

Roll Number: 101

City: Vijayawada

State: Andhra Pradesh

Program 14.2: Library Management

```
class Library {
    String libraryName;
    Library(String libraryName) {
        this.libraryName = libraryName;
    }
    class Book {
        String title;
        String author;
        Book(String title, String author) {
            this.title = title;
            this.author = author;
        }
        void display() {
            System.out.println("Library: " + libraryName);
            System.out.println("Book Title: " + title);
```

```
            System.out.println("Author: " + author);
        }
    }
}
public class LibraryManagement {
    public static void main(String[] args) {
        Library library = new Library("Central Library");
        Library.Book book =
                library.new Book("Java Programming", "James Gosling");
        book.display();
    }
}
```

Output:

Library: Central Library

Book Title: Java Programming

Author: James Gosling